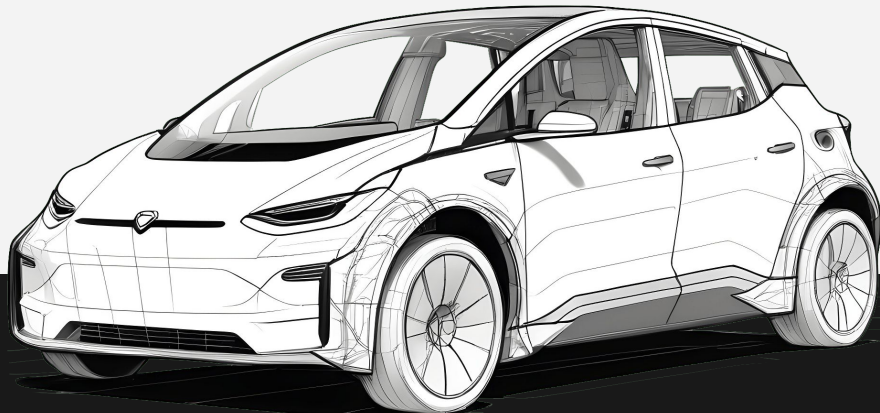


Used Car Characteristics and Price Prediction

An Exploratory and Predictive Data Analysis



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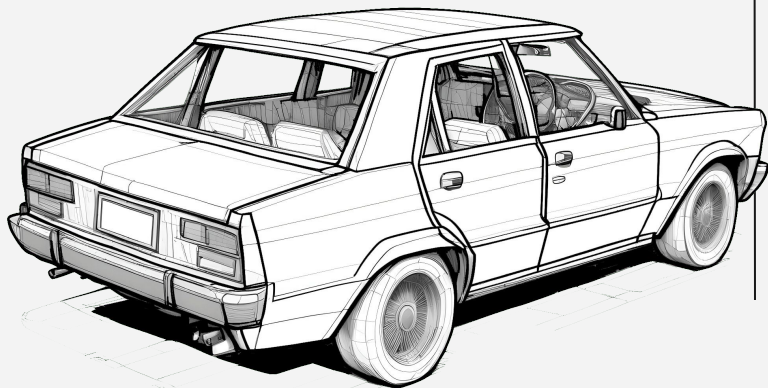
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Introduction, Scraping



Project Overview

What is This Project About?

- Analyze and predict used car prices using real-world online listings
- Focused on public data scraped from Autoscout24
- Built a complete data pipeline: scraping → cleaning → modeling
- Final goal: develop ML models that estimate fair prices based on car features

Why It Matters?

- Used car markets are opaque and inconsistent
- Buyers and sellers often lack pricing benchmarks
- Our system helps make pricing more data-driven and transparent

Raw Data Overview

- Source: AutoScout24
- Listings Scraped: 468 000
- Each listing contains 30+ features
- **“.json”** used to list links
- Stored in **“.parquet”** files
- Optimized for fast read/write during analysis and modeling

Scraped Feature Types

Numerical

Mostly common
descriptors of a car:
mileage, power, weight

Common attributes and
identification of car:
car_title, transmission

Categorical

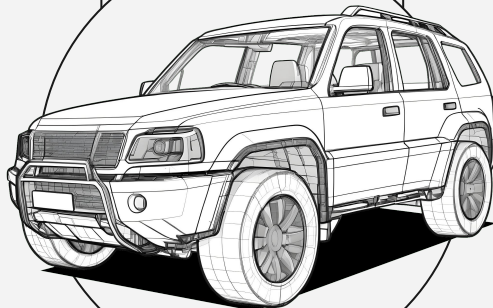


Composite

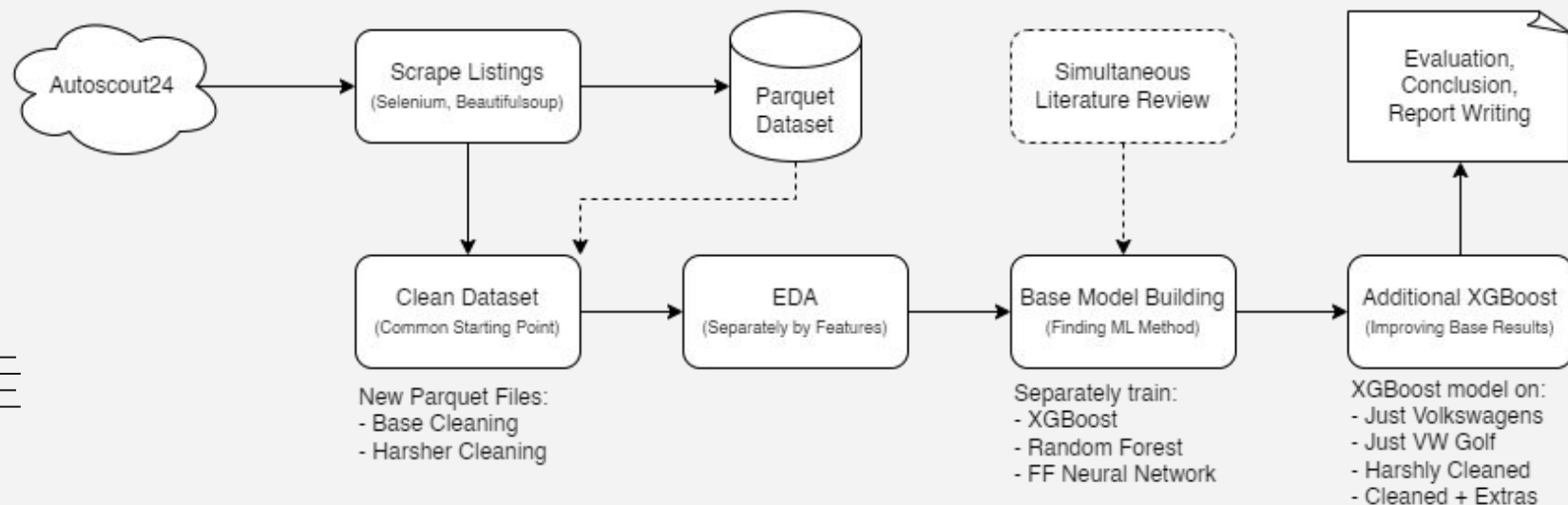
Composite ";" delimited
strings of features:
extras, safety_security

Features needing extra
transformation like:
first_registration -> age

Transformable



Project Workflow



Scraping Challenges and Solution

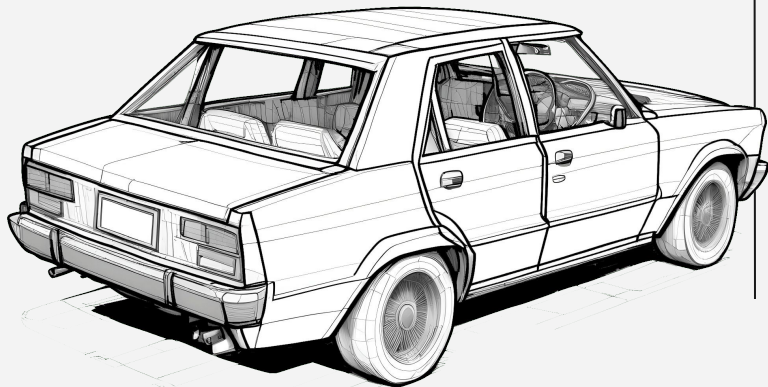
Challenges

- JavaScript-heavy pages → needed Selenium
- Frequent captchas and request blocking
- Slow runtime due to simulated user interaction
- Time pressure → scraping had to run in parallel
- Inconsistent listings & missing fields

Solution:

- Used Selenium + BeautifulSoup for dynamic content
- Added retry logic & modular structure
- Ran scraping in parallel (multi-core) to save time
- Extracted car brands dynamically
- Saved outputs in ".json" and ".parquet" for performance

02 EDA and Data Cleaning



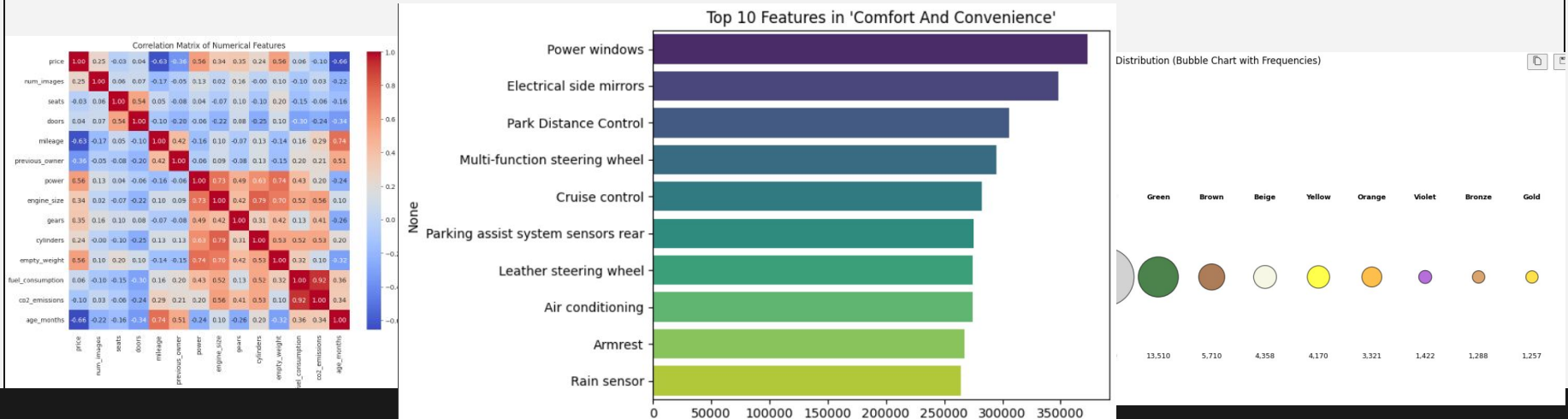
Our Mission

Research Questions:

1. How many features influence the price in a significant way?
2. Is it possible to accurately predict car costs from the features?

Sub-questions:

- What trends or patterns can we uncover by using simple EDA on car characteristics?
- Is it possible to process composite features in a way that helps us predict or rate cars?



Composite Features?

Composite features are columns in our dataset, where **each datapoint contains a long string**, with **additional features**

```
print(df["comfort_and_convenience"][10])
```

```
360° camera;Air conditioning;Air suspension;Armrest;Automatic climate control;Cruise control;Electrical side mirrors;Electrically adjustable seats;
```

Comfort and Convenience:

- Park distance control
- Cruise control

Entertainment and Media:

- Bluetooth
- Android Auto/Apple CarPlay

Safety and Security:

- ABS
- Driver-side airbag

Exploratory Data Analysis

Our main focus was to uncover as many interesting facts about used cars as we could, from the scraped data

Topics include:

- Overall details that describe the dataset
- Brand distributions
- Car dealerships vs private sellers
- Composite features
- Electric vehicle trends and patterns

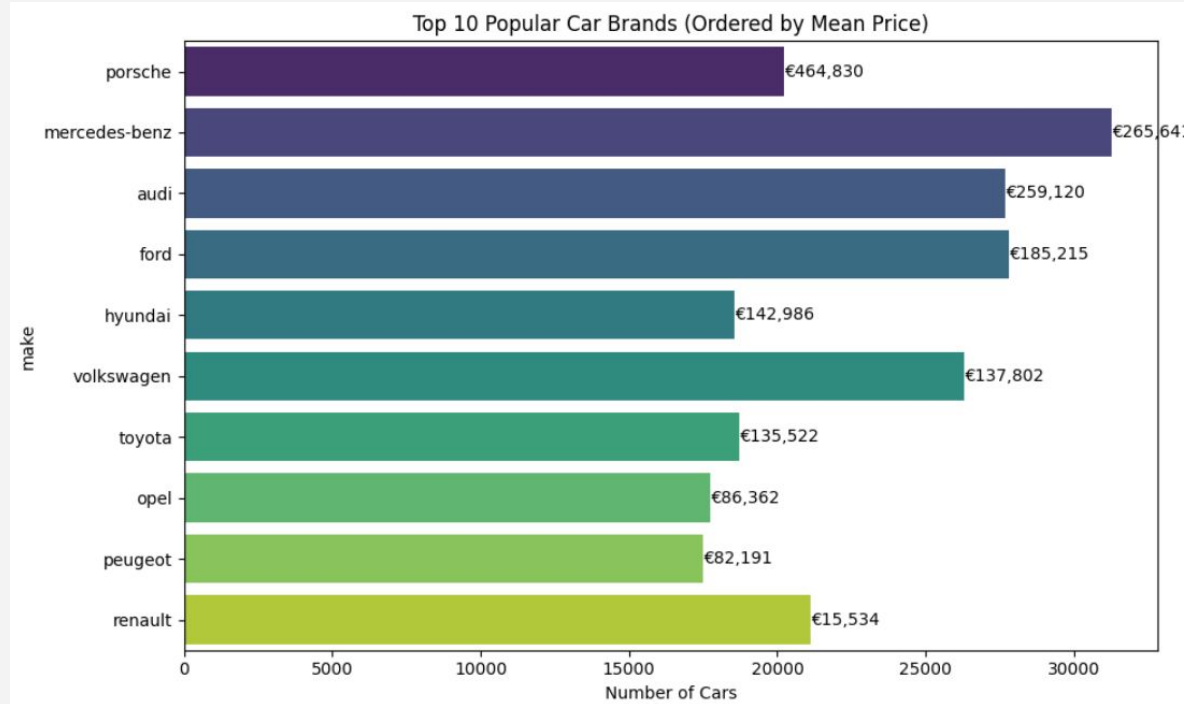
Overall Details

Summary statistics for numerical columns:

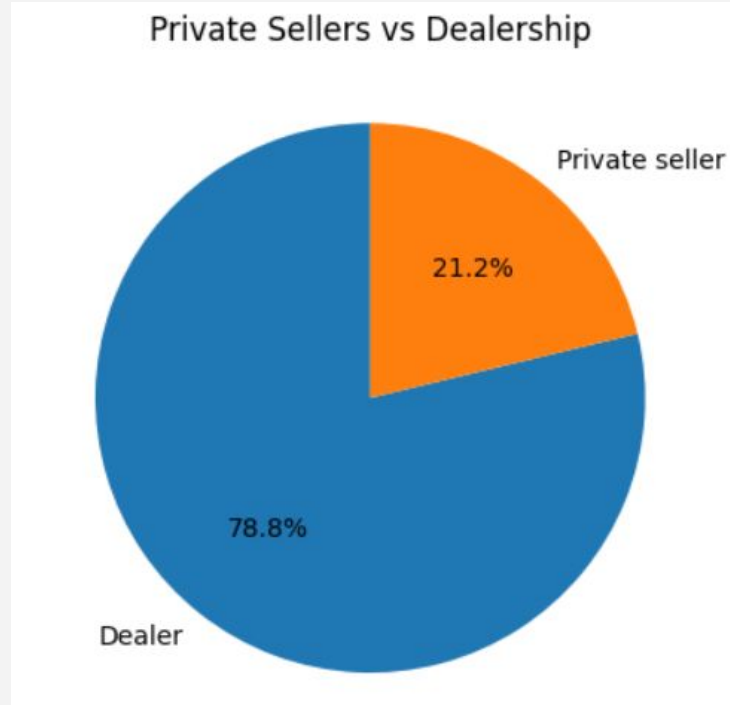
	price	num_images	mileage	previous_owner	power \
count	505615.00	499677.00	496658.00	224209.00	504847.00
mean	133239.14	16.16	89259.09	1.51	198.03
std	896730.36	6.28	359666.93	1.20	134.63
min	1.00	2.00	0.00	1.00	1.00
25%	10500.00	14.00	19500.00	1.00	109.00
50%	24950.00	17.00	70975.50	1.00	158.00
75%	114897.50	17.00	135949.75	2.00	241.00
max	55555555.00	53.00	244000035.00	99.00	999.00

	engine_size	empty_weight	co2_emissions
count	431989.00	361763.00	299130.00
mean	2027.44	1584.80	409.45
std	1144.05	497.01	51818.21
min	0.00	1.00	0.00
25%	1397.00	1263.00	49.00
50%	1798.00	1530.00	130.00
75%	2198.00	1860.00	170.00
max	99999.00	61290.00	9999999.00

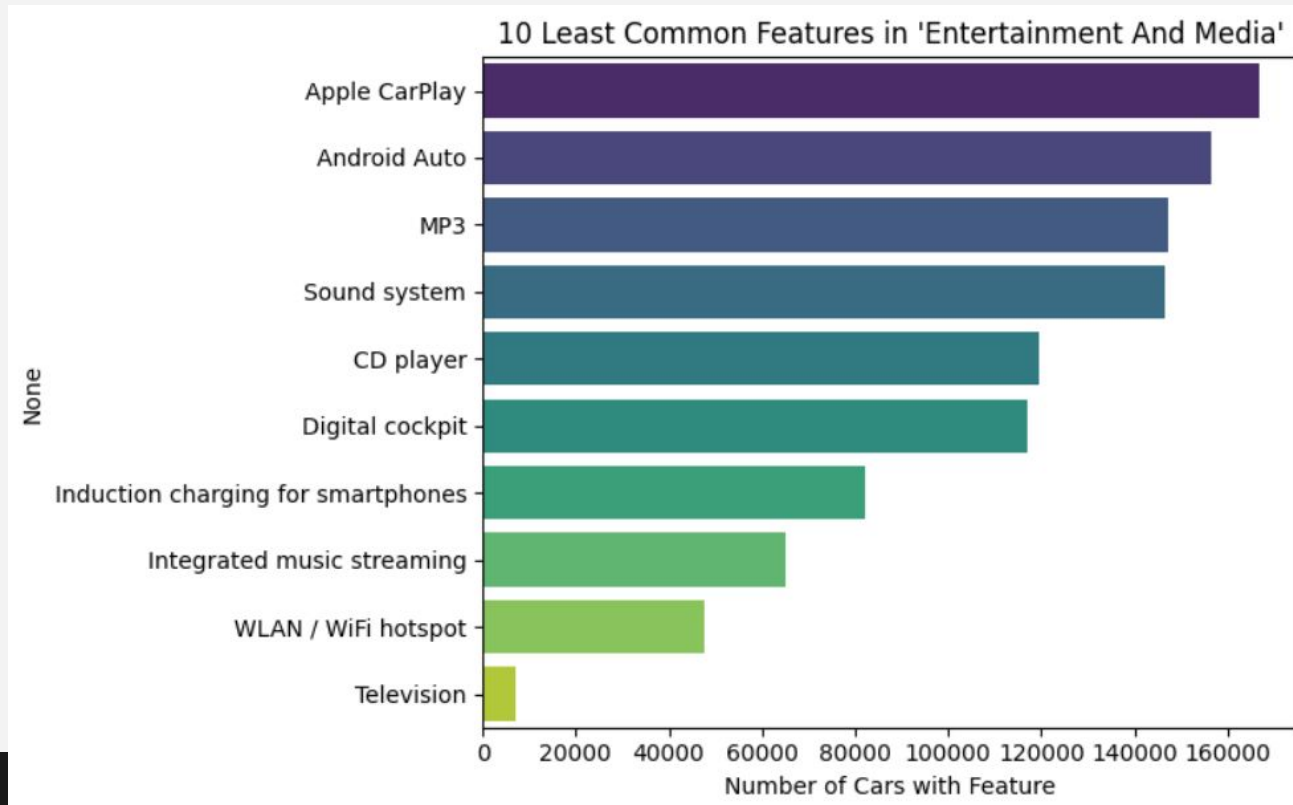
Brand Distributions



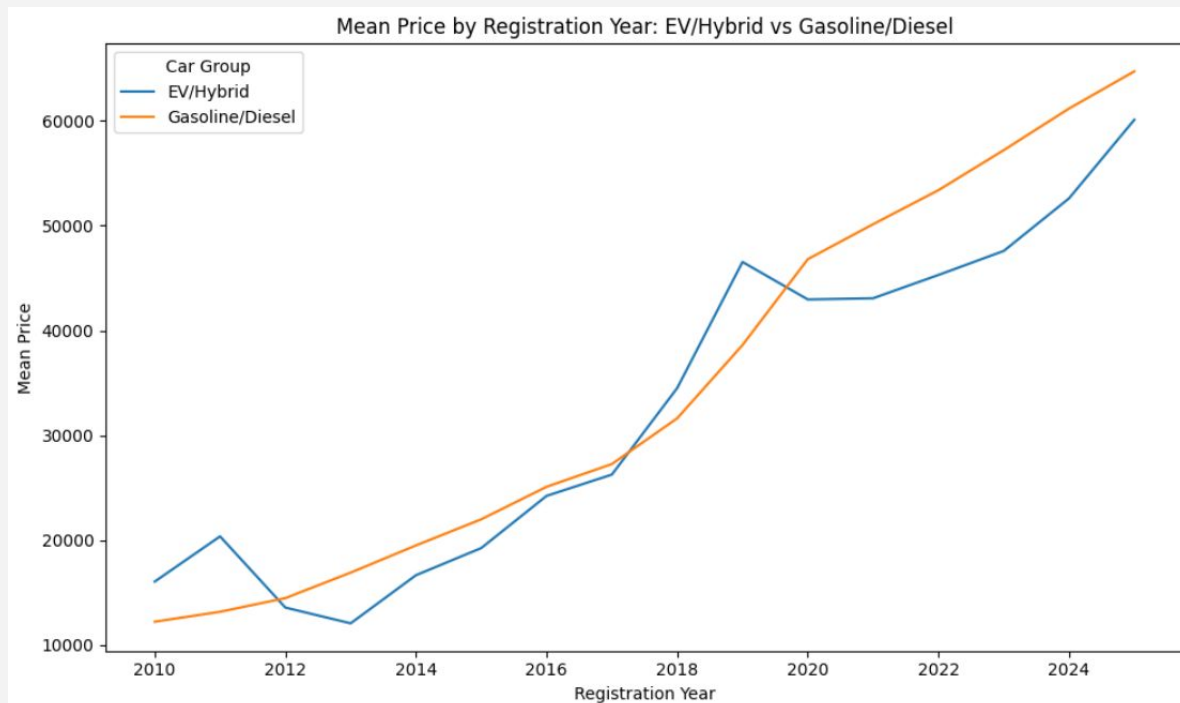
Dealerships vs Private Sellers



Composite Features



EV Trends



Data Cleaning

Flowchart of Cleaning

1. – Raw Data



- Raw Data, directly scraped from Autoscout24

2. – Basic Cleaning



- Cleaning for Outliers, NAs and duplicates
- Also translated some (first_registration → age)

(3. – For some models) Harsh Cleaning

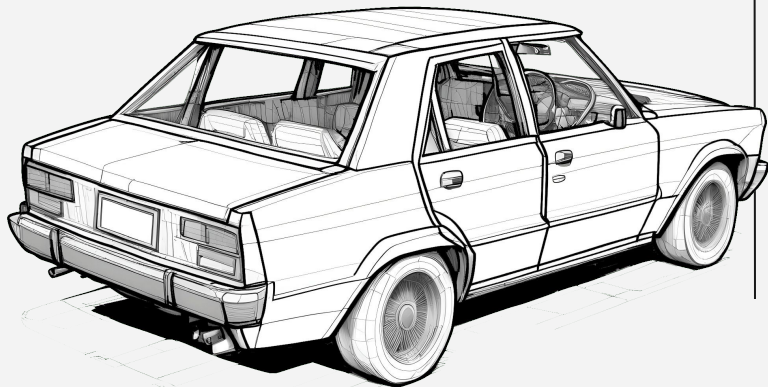


- Harsher cleaning: for outliers with stricter ranges
- Stricter caps, e.g. price, fuel consumption etc..., so we filter for rare vehicles too

03

Predictions, Literature

And our conclusion from them



Literature Review

Findings

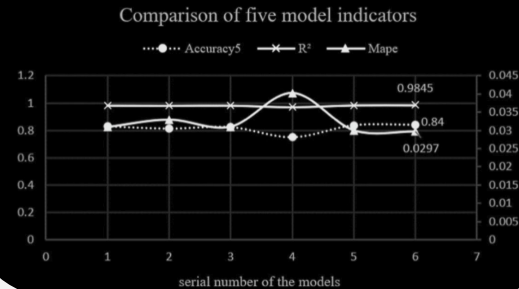
Having inspected and cited more than 10 scientific papers, we have found that the literature is very interested in efficient car price predictions.

Usually these papers used several, different models on either scraped, primary datasets or secondary data from sources like Kaggle.

The best models had an R^2 range of 0.85 – 0.98, with a multitude of different models.

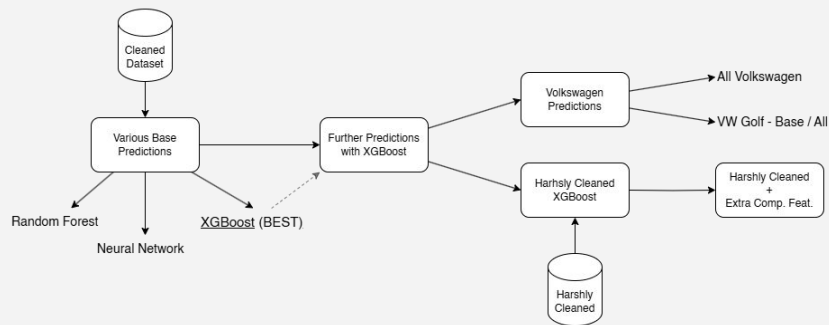
Huang et al. 2022 [1]

	Mape	Accuracy ₅	R ²
XGBoost	0.0309	0.8283	0.9827
CatBoost	0.0329	0.8118	0.9814
LightGBM	0.0310	0.8228	0.9828
ANN	0.0402	0.7486	0.9713



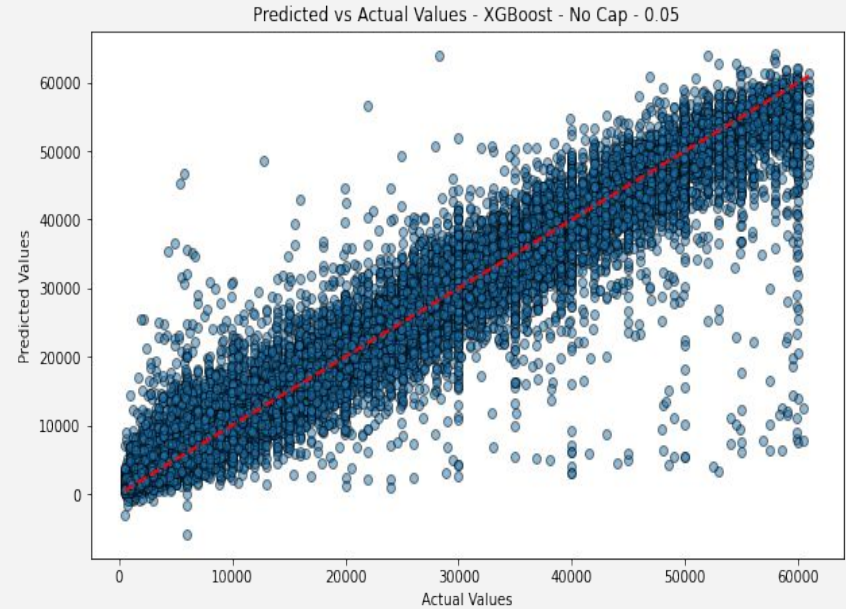
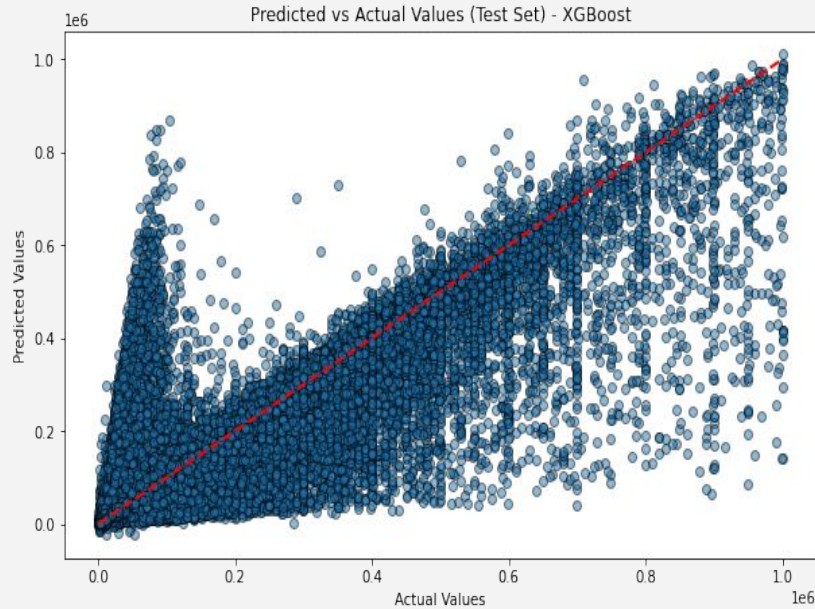
Our Predictions

Method	Mean Absolute Error (MAE)	R-squared (R^2)
XGBoost	29770.0580	0.7949
Random Forest	29822.8709	0.7823
Neural Network	34476.8203	0.7601
XGBoost (Volkswagen)	35747.3092	0.8295
XGBoost (All VW Golfs)	9015.9600	0.8768
XGBoost (Base VW Golfs)	2035.6000	0.8936
XGBoost (Harshly Cleaned)	1806.8305	0.9439
XGBoost (Harshly Cleaned Extras)	1735.6964	0.9496



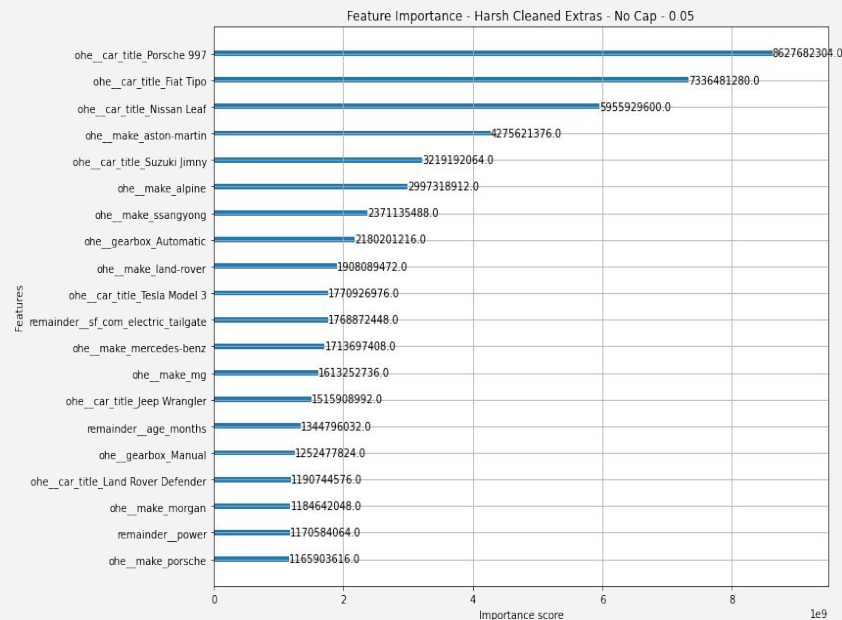
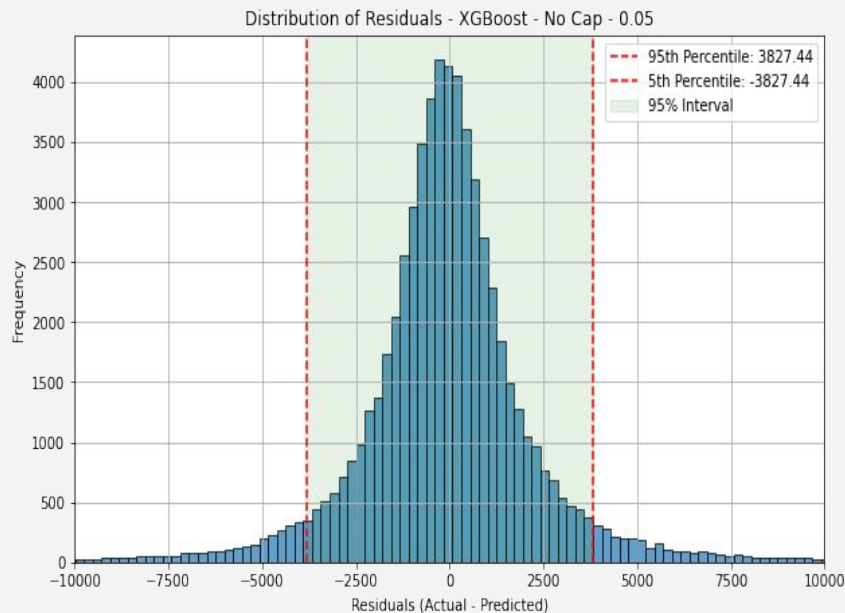
- Multi-step approach, first we found out what the best prediction model is, which in our case was XGBoost, then iteratively tried to get a better result by filtering / cleaning the data.
- The base models had an okay performance with R score of almost 0.8, but looking at the MAE, it made pretty big errors even considering the 1 million euro max cap. Not suitable for production.
- With just VW, the results improved a little, the real improvements came after further filtering to Golfs.
- After a much harsher cleaning we managed to approach the results of the literature with an R2 of 0.95 and MAPE of 16%, 1735 MAE. + Composite features yielded marginal improvement.

Predicted vs Actual Plots (Comparison)



Significantly less underestimation, no protrusion under 100k.
Note the different scales.

Additional Plots (Best model)

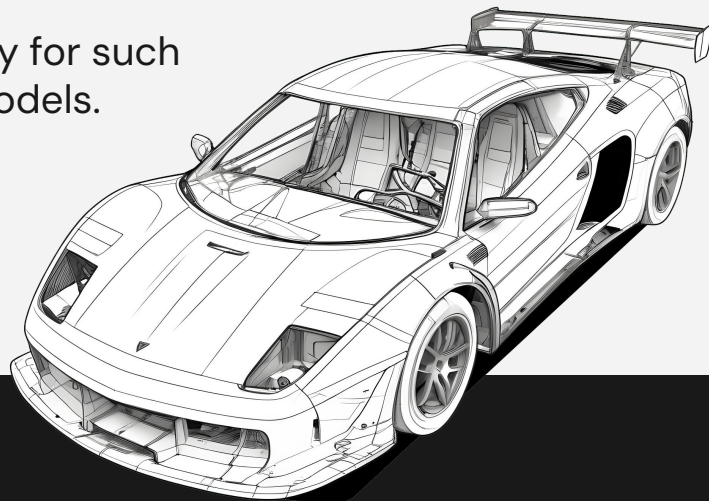


The residuals have a pretty nice distribution and quite a sharp peak.
Feature importance is as expected, mainly base stats + car_title / brands.

Final Thoughts

After all, we can conclude the following:

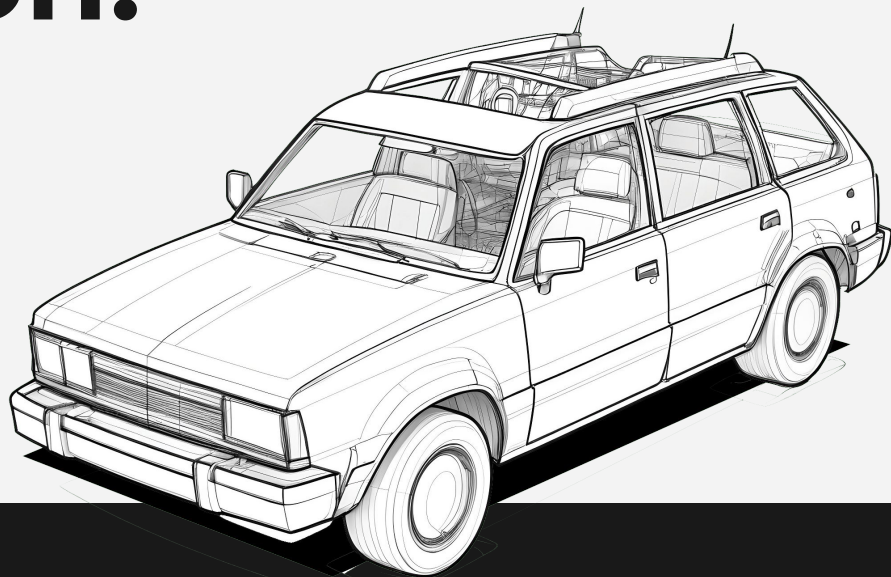
- It is very hard, likely impossible to build one singular prediction model for a whole marketplace of cars with great range.
- There needs to be a comprehensive strategy for such predictions, e.g. by price range and make models.
- The scientific literature in the topic, while numerous, it is not very helpful, as often not much is disclosed of the data.



Thank you for your attention!

Created for the SS/25 Data Science
Capstone Project at IMC Krems by:

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Márió Palágyi



Citations, Resources

Citations

[1] J. Huang, Z. Yu, Z. Ning, and D. Hu, "Used Car Price Prediction Analysis Based on Machine Learning," presented at the 2022 International Conference on Artificial Intelligence, Internet and Digital Economy (ICAID 2022), Atlantis Press, Dec. 2, 2022, pp. 356–364, ISBN: 978-94-6463-010-7. DOI: 10.2991/978-94-6463-010-7_37. [Online]. Available: <https://www.atlantis-press.com/proceedings/icaid-22/125977128> (visited on 05/20/2025).

Resources

Our public repository with data can be found [here on GitHub](#).

Presentation template by [Slidesgo](#).

